

Experiment 17

Title

Write a Prolog Program to Sum the Integers from 1 to n.

Aim

To develop a Prolog program to find the sum of integers from **1 to n** using recursion.

Objective

To understand the concept of recursion in Prolog by computing the sum of natural numbers from **1 to n**, using a base case and a recursive rule.

Algorithm

Step 1: Define the base case where the sum of 0 is 0.

Step 2: Accept a positive integer **N**.

Step 3: Decrease **N** by 1.

Step 4: Recursively calculate the sum from **1 to N-1**.

Step 5: Add **N** to the previous sum.

Step 6: Display the final sum.

Program

% Base Case

sum(0,0).

% Recursive Case

sum(N,Sum) :-

 N > 0,

 N1 is N - 1,

 sum(N1,Sum1),

 Sum is Sum1 + N.

Sample Input

?- sum(5,Sum).

Sample Output

```
GNU Prolog console
File Edit Terminal Prolog Help
GNU Prolog 1.5.0 (64 bits)
Compiled Jul  8 2021, 12:33:56 with cl
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| ?- consult('C:/Users/B SANTHOSH KUMAR/OneDrive/Documents/exp17.pl').
compiling C:/Users/B SANTHOSH KUMAR/OneDrive/Documents/exp17.pl for byte code...
C:/Users/B SANTHOSH KUMAR/OneDrive/Documents/exp17.pl compiled, 6 lines read - 850 bytes written, 26 ms

yes|
| ?- sum(5, Sum).

Sum = 15 ?

(125 ms) yes
| ?-
```

Result

Thus, the Prolog program to find the **sum of integers from 1 to n** using recursion was executed successfully, and the correct sum was obtained.